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Procedia Computer Science 00 (2025) 000–000

Procedia Computer Science

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International Conference on Machine Learning and Data Engineering

Synergistic Feature Fusion Approach Towards Real-Time Children Facial Emotion Recognition

Shivaprasad D L[a]* and D S Guru[a]

[a]*Department of Studies in Computer Science, University of Mysore, Manasagangotri, Mysuru - 570006, India.*

Abstract

Accurate recognition of children's facial emotions remains challenging due to subtle expression variations, dynamic pose changes, and the scarcity of appropriate datasets. To address these issues, this study introduces a newly developed children's facial emotion dataset encompassing six primary expressions across multiple orientations. Leveraging this resource, a video-based emotion recognition framework is proposed, which combines key-frame extraction, a customised YOLOv5 face detector, and a synergistic feature fusion strategy integrating deep semantic features from EfficientNetB0 with handcrafted HOG descriptors. Feature dimensionality is optimised via Linear Discriminant Analysis (LDA), and classification is performed using a Support Vector Machine (SVM) with an RBF kernel. Extensive experiments on 8,234 frames demonstrate that the proposed framework achieves 98.84% accuracy, consistently outperforming baseline methods, including FEAT, DeepFace, and FER, as well as alternative feature representations derived from other deep networks (AlexNet, GoogleNet, MobileNet, ResNet, and VGG19), and conventional LBP descriptors. These results highlight the effectiveness of synergistic feature integration and the importance of the curated dataset, establishing the framework as a robust solution for real-time applications in education and healthcare.

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Peer-review under responsibility of the scientific committee of the International Conference on Machine Learning and Data Engineering

Keywords: Key-frame Extraction; Synergistic Feature Fusion; Children Expression Recognition.

1. Introduction

Human emotions are central to non-verbal communication, functioning as an implicit yet powerful channel for expressing thoughts, intentions, and psychological states [1]. Among the many modalities of emotional expression,

* Corresponding author. Tel.: +91-9741406556.
E-mail address: shivaprasaddl143@gmail.com; dsg@compsci.uni-mysore.ac.in.

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facial cues are widely regarded as the most natural and universally interpretable indicators of affective states [2]. The automatic recognition of such cues has become an important focus in human–computer interaction, as it enables computational systems to better understand and respond to human behaviour. While significant research has been devoted to emotion recognition in adults, the study of children's emotional expressions has gained attention only recently. This area presents unique challenges because children often display emotions in subtle, spontaneous, and highly dynamic ways compared to adults [3]. Developing reliable systems for identifying children's affective states is therefore a non-trivial task, yet one with high practical relevance. Accurate detection of children's emotions can support parents in nurturing and responding to their child's needs, assist educators in tailoring instructional strategies for improved classroom engagement, and provide paediatricians and psychologists with valuable insights into emotional well-being for diagnosis and therapeutic interventions [4][5][6]. In real-world contexts, emotional understanding typically relies on a combination of verbal and non-verbal cues. Within computational frameworks, however, vision-based methods focusing on facial expression analysis have become predominant due to advances in computer vision and deep learning [7]. Despite the progress in adult emotion recognition, work directed specifically toward children remains limited. One major bottleneck is the scarcity of publicly available datasets that capture the variability of children's expressions across diverse poses, environments, and cultural settings [8]. The absence of such resources constrains the development of generalisable and robust recognition models. To address this gap, the present study introduces a comprehensive dataset of children's facial expressions that incorporates a wide range of poses and contexts. Building on this dataset, we propose a synergistic feature-fusion framework that combines convolutional features with handcrafted descriptors in a unified architecture. This hybrid approach aims to improve recognition accuracy and robustness in real-time scenarios. Beyond methodological contributions, the proposed system holds promise for deployment in educational, domestic, and healthcare applications, where reliable understanding of children's emotions can create tangible benefits. The main contributions of this work are summarised as follows:

- Developing a large-scale children's facial emotion dataset encompassing diverse poses, expressions, and contexts.
- Designing a novel synergistic feature-fusion model integrating deep convolutional features with handcrafted descriptors for robust classification.
- Extensive evaluation demonstrating improved accuracy and generalisability compared to existing approaches.

The structure of this paper is as follows: Section 2 reviews the existing literature on expression recognition. Section 3 presents the proposed real-time system for children's expression recognition, while Section 4 discusses the developed children's emotion dataset in detail. Section 5 reports the experimental results along with a discussion, and Section 6 concludes the study.

2. Literature Study

Real-time emotion prediction has emerged as a crucial area of research due to its broad applications in healthcare, education, behavioural sciences, and human–computer interaction [1]. Several modalities have been explored for this task, including facial expressions, acoustic features from speech, physiological measures such as galvanic skin response (GSR), and neurophysiological signals like electroencephalography (EEG) [2]. Among these, vision-based facial expression analysis remains the most widely studied because it is non-invasive and can be implemented with commonly available imaging devices [3]. Research on facial expression recognition (FER) has primarily evolved along three principal directions. The earliest works employed handcrafted feature extraction techniques, where texture and gradient-based descriptors such as Local Binary Patterns (LBP) [4], Histogram of Oriented Gradients (HOG), and Wavelet Gradient Transform [5] were applied to encode facial details. These handcrafted features were often coupled with classifiers such as Support Vector Machines (SVM) or k-Nearest Neighbours (k-NN). While such methods are computationally efficient, they typically perform poorly under uncontrolled conditions, such as changes in illumination, pose variations, and occlusions. The second research stream is led by deep learning approaches, particularly Convolutional Neural Networks (CNNs), which learn hierarchical feature representations directly from raw images. Networks such as VGG-19, ResNet-50, and Inception-V3 have become standard benchmarks for FER tasks [6][7]8]. These architectures outperform traditional methods in accuracy but demand large annotated datasets and considerable computational resources. More recently, data augmentation and generative strategies, including conditional Generative Adversarial Networks (cGANs), have been employed to enrich training data and improve

robustness in scenarios with limited samples [9]. The third line of work focuses on hybrid or synergistic frameworks, which integrate handcrafted and deep features to exploit their complementary strengths. Several studies report that combining CNN-based embeddings with classical descriptors leads to improved generalisation under real-world conditions [10][11][12]. Transfer learning techniques, where pre-trained deep features are classified using SVMs, Random Forests, or k-NN, have also been widely applied [13][14][15]. Such strategies are particularly beneficial when working with smaller datasets, as they reduce overfitting and enhance robustness. Additionally, landmark-based methods that track facial keypoints (e.g., corners of the mouth, eyebrows, and eyes) have been explored for lightweight real-time applications [16][17]. However, these approaches are highly sensitive to occlusion and variations in head pose. A further direction is multimodal emotion recognition, where facial data are fused with other signals such as speech, EEG, or GSR to capture richer affective representations [18][19][20]. These multimodal frameworks generally achieve higher accuracy and reliability, though they also introduce complexity in data acquisition and integration. Despite these advances, a consistent limitation across the literature is the scarcity of high-quality datasets that accurately represent children's facial expressions. Widely used datasets such as CK+ [21], FER2013 [22], RAF-DB [23], and AffectNet [24] primarily focus on adult subjects. A few resources do include children, for example, the Dartmouth Database of Children's Faces [25]; however, such datasets typically contain only frontal facial images acquired under controlled settings. As a result, they fail to capture the natural variability, spontaneity, and subtlety of children's emotional expressions in unconstrained environments. This gap significantly restricts the development of robust models that generalise well to real-world child-centric applications. To overcome this limitation, the present study introduces a new dataset of children's facial expressions that covers a broad age range (4–16 years) and incorporates multiple pose orientations and environmental conditions. Table 1 provides a comparative summary of widely used FER datasets alongside our proposed dataset, highlighting the differences in scale, diversity, and applicability. Based on this resource, we design a synergistic feature-fusion framework that integrates convolutional deep features with handcrafted descriptors to enhance classification robustness and real-time applicability. Through this contribution, our work advances child-specific emotion recognition research and provides a foundation for the practical deployment of emotion recognition in education, healthcare, and domestic contexts.

Table 1. Overview of Existing Datasets and the Proposed Children's Emotion Dataset.

Dataset	Participants	Facial pose angle	Strengths	Limitations
CK+ (Extended Cohn–Kanade) [21]	Adults	Frontal	Standard benchmark, FACS-coded expressions	Adult-only, constrained environment
FER2013 [22]	Adults	Frontal	Large scale, varied capture conditions	Low resolution, very few child samples
RAF-DB [23]	Adults	Frontal	Annotated with basic and compound emotions	Focuses on adults, limited age diversity
AffectNet [24]	Adults	Frontal	Extensive dataset, diverse expressions	Sparse child data, unbalanced labels
Dartmouth Children's Faces [25]	Children (5-12 years)	Frontal	One of few child-focused datasets	Only frontal poses, controlled setting
EmoReact [26]	Children (2-8 years)	Video (Face + Audio)	Naturalistic expressions from online sources	Small dataset, noisy labelling
Proposed	**4-16 years (360 videos, 8,324 images)**	**0 – 90 Orientation**	**60 30 male & 30 female**	**Only children faces with different facial poses**

3. Proposed Framework

This research presents a video-based framework for predicting children's facial expressions. The system functions in two main phases: training and testing, with the overall workflow illustrated in Figure 1. The framework is structured into five core modules: data collection, key-frame selection, synergistic feature fusion, feature transformation, and expression identification. Each module serves a specific purpose within the pipeline and is elaborated in the subsequent subsections.

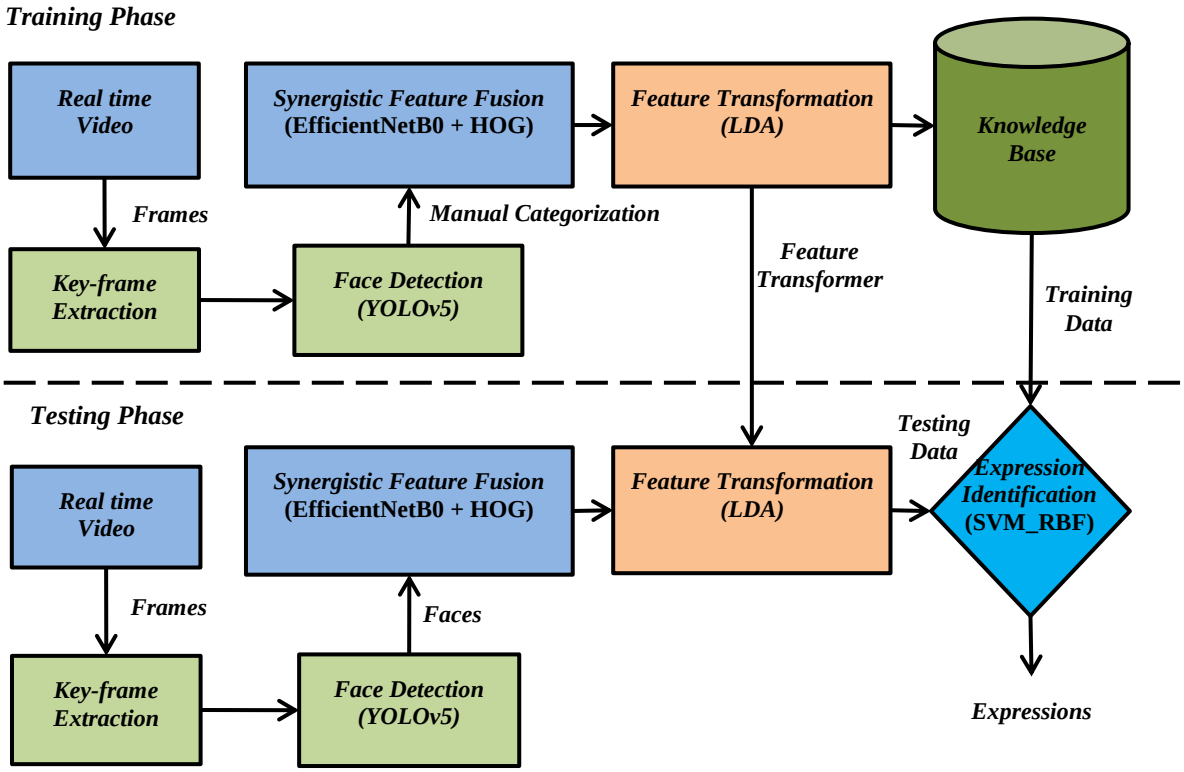


Fig.1. Demonstration of the proposed framework.

3.1. Data Collection

Data collection refers to the process of collecting raw input for the system. In this work, emotional responses of children were recorded in real-world environments using multiple camera setups. The collected video sequences served as the primary source for building the training dataset. In contrast, live video streams were employed during the testing phase to evaluate the system's real-time performance.

3.2. Keyframe Selection

Once the video is captured, the next step involves keyframe selection, which is carried out using the proposed mechanism outlined below:

- All frames (F) are first collected from the input video.
- For each frame, histograms of the three color channels, Red, Green, and Blue are computed (HF).
- A similarity matrix (SF) is then generated by comparing the histograms across frames.
- Based on the similarity matrix and a predefined threshold (th), visually similar frames are grouped together.
- Within each group, the frame exhibiting the maximum gradient value is chosen, as the gradient reflects the sharpness and overall quality of the frame.
- The chosen frames are designated as key frames and are forwarded to subsequent stages of the system.

This approach ensures that only distinctive and high-quality frames are retained for both the training and testing phases. Importantly, during the testing stage, the extraction process is executed within a minimal time window, thereby enabling real-time applicability.

3.3. Face Detection

Following key-frame selection, the facial regions are isolated using a dedicated face detection module. In this work, the widely recognised YOLOv5 (You Only Look Once, version 5) object detection framework was adapted and fine-tuned on the proposed dataset of children's faces, thereby developing a specialised face detector. The choice of YOLOv5 is significant, as it offers a strong balance between detection accuracy and computational efficiency, making it highly suitable for real-time applications. Unlike conventional face detectors, which often struggle with poor image quality, varied lighting, and non-frontal poses, the customised YOLOv5 model demonstrates robustness in handling diverse facial orientations and challenging conditions. This ensures reliable detection of children's faces, which is a crucial step for accurate downstream emotion recognition.

3.4. Synergistic Feature Fusion

Once the facial regions are accurately localised from each key-frame, the next stage focuses on feature extraction, which is need for achieving reliable and discriminative recognition. In this work, a dual strategy is employed by integrating both deep learning-based and handcrafted approaches. Specifically, EfficientNetB0, a lightweight yet highly effective convolutional neural network, is adopted as the deep feature extractor. EfficientNetB0 is designed using a compound scaling principle, which balances network depth, width, and resolution to achieve state-of-the-art accuracy with reduced computational cost. This makes it particularly suitable for real-time applications, where both performance and efficiency are critical. In parallel, handcrafted descriptors are obtained using the Histogram of Oriented Gradients (HOG) technique. HOG is well-suited for capturing local edge orientations and contour structures, thereby providing a robust geometric description of facial regions. While EfficientNetB0 captures high-level semantic representations that reflect subtle emotional variations, HOG emphasises structural and shape-based features, which remain effective under varying illumination and pose conditions.To fully exploit the complementary strengths of these methods, the extracted features are fused into a single representation. For each image, EfficientNetB0 generates **m** features, while HOG produces **k** features. These are combined into a unified feature vector: $\mathbf{F_v} = \mathbf{F_{EfficientNetB0}} \cup \mathbf{F_{HOG}} = \{\mathbf{g_1, g_2, \ldots, g_m}\} \cup \{\mathbf{h_1, h_2, \ldots, h_k}\}$. This enriched feature space enhances both the accuracy and the robustness of the classification process by leveraging deep semantic cues alongside precise geometric descriptors.

In addition to the deep and handcrafted features, two supplementary attributes are incorporated to further enrich the representation. Specifically, the coordinate values of the detected face along the x- and y-directions are included, which provide crucial spatial information regarding the position and orientation of the face within the frame. These coordinates are extracted using the MediaPipe library [27], a widely adopted toolkit for real-time face landmark detection and pose estimation. By integrating these positional cues, the framework is able to account for face angle variations and subtle pose shifts, thereby improving robustness in dynamic video scenarios. With this enhancement, the final feature vector attains a dimensionality of **(m+k+2)**, where **m** corresponds to the number of EfficientNetB0 features, **k** to the number of HOG descriptors, and the additional two represent the spatial coordinates. A sample of face angle estimation is illustrated in Figure 2. This augmented feature space ensures that the system captures not only semantic and structural information but also geometric and positional variations, leading to a more comprehensive representation for expression classification.

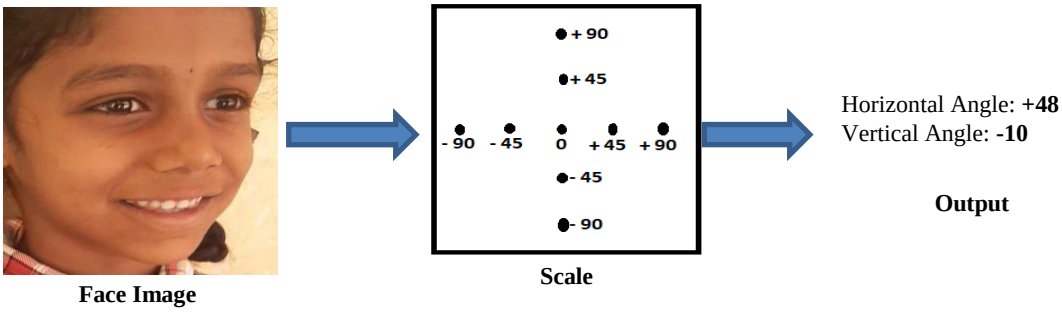


Fig. 2. Computation of facial pose orientation angles.

3.5. Feature Transformation

After extracting the synergistic features, the resulting high-dimensional space is further refined through transformation-based dimensionality reduction. In this study, Linear Discriminant Analysis (LDA) is employed, as it is particularly well-suited for classification-oriented tasks. Unlike unsupervised techniques, LDA leverages class label information to project the original features into a lower-dimensional subspace that maximises inter-class separability while simultaneously minimising intra-class variability. This characteristic makes it highly effective for strengthening the discriminative capacity of the system. Let **Fv** represent the concatenated feature vector from the previous stage, containing $\mathbf{m + k + 2}$ dimensions. Through LDA, this vector is transformed into a compact representation **TFv** with **d** dimensions, where $\mathbf{d << (m + k + 2)}$, expressed as: $\mathbf{TF_v = \{p_1, p_2, \ldots, p_d\}}$. This dimensionality reduction not only retains the most salient and class-relevant attributes of the data but also improves computational efficiency, thereby accelerating real-time performance without compromising accuracy.

3.6. Expression Identification

During the training phase, the input video data is initially processed to extract key frames, from which facial regions are identified using the proposed face detection model. The extracted face images are then manually labeled into six predefined emotion classes: Angry, Disgusted, Happy, Neutral, Sad, and Surprised. Following annotation, synergistic features are extracted through the combined deep and handcrafted approaches, and their dimensionality is reduced using the transformation method described earlier. The resulting compact features are then used to train a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel, and the learned model is stored within a structured knowledge base. The SVM-RBF was chosen for its ability to handle non-linear decision boundaries and high-dimensional feature spaces effectively, making it particularly suitable for emotion recognition tasks where class separability is often complex and subtle. In the testing phase, the same pipeline is executed, except for manual labelling. The trained SVM-RBF model classifies the extracted features into the appropriate emotion categories. To ensure a reliable evaluation, the system's performance was assessed across T independent trials using a fixed train–test split. At the same time, multiple rounds of cross-validation were also conducted to confirm robustness further. Detailed logs of each trial were maintained, and the best-performing combination of dimensionality reduction and classification strategy was preserved in a dedicated knowledge repository for future deployment. By replicating the complete training workflow during testing, the system guarantees consistency, reproducibility, and stability in real-time classification outcomes.

4. Dataset Description

For the experimental study, we constructed a dedicated video dataset of children's facial expressions through recordings collected in naturalistic environments such as schools, parks, and care centres. Participation was entirely voluntary, and explicit permission was obtained from parents or legal guardians before recording, ensuring compliance with ethical standards for research involving minors. The dataset includes 60 children (30 boys and 30 girls) within the age range of 4 to 16 years. Each participant contributed six video sequences corresponding to the six target emotional categories: Angry, Disgusted, Happy, Neutral, Sad, and Surprised. To simulate realistic conditions, the recordings were designed to capture diverse pose orientations rather than restricting the data to frontal views. This design choice makes the dataset more challenging and representative of natural child behaviour. From the video corpus, 8,234 representative frames were extracted using the proposed keyframe selection strategy. The extracted face regions were then preprocessed using the customised face detector. Figure 3 illustrates sample frames from the dataset, while Figure 4 shows examples of pose variations after preprocessing. The gender distribution of participants is provided in Figure 5, and pose-angle statistics are summarised in Table 2. While the dataset represents a meaningful step toward advancing research in child-specific facial expression recognition, we recognise that it is region-specific and may not fully capture cross-cultural variability in emotional display. Cultural and environmental factors can influence subtle expression patterns, which introduces the possibility of bias. To address this limitation, we plan to expand the dataset in future work by including children from different cultural and socio-demographic backgrounds. Making the dataset more broadly accessible, subject to ethical guidelines, is also part of our long-term vision to encourage reproducibility and comparative research within the community.

Angry **Disgust** **Happy** **Neutral** **Sad** **Surprise**

Fig.3. Example samples of the dataset.

Up - 45⁰ **Right - 90⁰** **Right - 45⁰** **Frontal** **Left - 45⁰** **Left - 90⁰** **Down - 45⁰**

Fig.4. Different pose orientation of the Happy expression class.

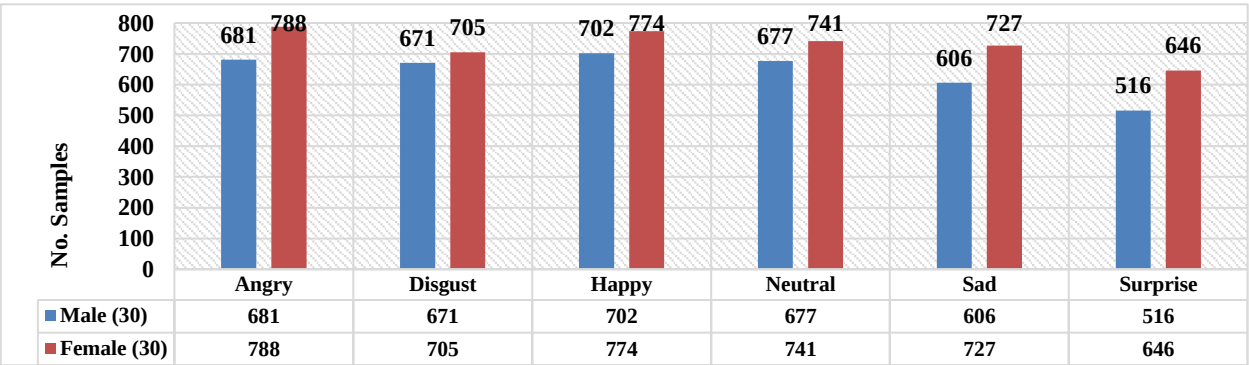


Fig. 5. Gender wise statistics of the created emotion dataset.

Table 2. Facial pose wise statistics of the created dataset.

Expression	Number of Samples								
	0^0 to 5^0		0^0 to 10^0		0^0 to 45^0		0^0 to 90^0		
	Left	Right	Left	Right	Left	Right	Left	Right	Total
Angry	272	238	462	362	914	547	916	553	**1469**
Disgust	308	265	480	429	663	696	667	709	**1376**
Happy	312	349	409	555	497	961	504	972	**1476**
Neutral	299	308	389	500	473	927	479	939	**1418**
Sad	306	244	421	427	605	719	608	725	**1333**
Surprise	157	144	244	257	359	703	394	768	**1162**
Total	**1654**	**1548**	**2405**	**2530**	**3511**	**4553**	**3568**	**4666**	**8234**

5. Experimental Analysis

5.1. Experimental Setup

This section outlines the experimental setup and presents the analysis of the obtained results. To evaluate the performance of the proposed framework, we utilised the children's emotion dataset, which comprises 8,234 images with diverse facial pose orientations. As a preparatory step, 2,000 images covering a wide range of orientations were manually annotated to mark the facial regions. From this annotated subset, 1,500 samples were used to fine-tune the YOLOv5 model, enabling the development of a domain-specific and robust face detector. Conventional face detection methods were observed to be inadequate in this context, as they frequently misclassified non-facial regions or failed to detect faces under extreme conditions such as side profiles or upward-tilted views. To illustrate these limitations, the proposed YOLOv5-based detector was compared with widely used alternatives, including MTCNN [3], Img2Pose–C [19], and FER [22]. The comparative outcomes are summarised in Figure 6. The remaining 500 annotated samples were reserved for validation. On this set, the proposed detector achieved a mean Average Precision (mAP) of 85%, demonstrating its effectiveness in handling challenging orientations. From the comparative analysis, we observed that although MTCNN performed better than other existing detectors, it still failed in cases involving upward-facing or near 90-degree profiles. However, the fine-tuned YOLOv5 detector consistently recognised faces across a wide range of poses, underscoring its suitability for real-world child-specific emotion recognition. Following successful face detection, synergistic feature extraction was performed as detailed in Section 3.3. Two complementary feature types were utilised: (i) deep features generated by the EfficientNetB0 model, and (ii) handcrafted features derived from the Histogram of Oriented Gradients (HOG). EfficientNetB0 was adapted by modifying its final layer to produce 1,000 discriminative features. In contrast, the HOG descriptor, configured for 256 × 256-pixel images with eight orientations, a 32 × 32 cell size, and a 2 × 2 block structure, yielded 1,568 features per image.

The feature extraction process yielded two matrices: one of size 6,052 × 1,000 from EfficientNetB0 outputs and another of size 6,052 × 1,568 from HOG descriptors, along with two extra coordinate attributes. These were concatenated to form a combined matrix of size 6,052 × 2,570, which was normalised to the 0–1 range using Standard Scaler preprocessing. For classification, the dataset was split into training and testing subsets in a 70:30 ratio, maintaining balanced representation across all emotion classes. To ensure statistical reliability, the experiments were repeated over ten independent trials with cross-validation (T = 10), where the test samples were unseen in each trial. Dimensionality reduction was then applied using Linear Discriminant Analysis (LDA), reducing the 2,570-dimensional feature space to five discriminative components while preserving the features most relevant for class separation. The resulting reduced feature set was classified using a Support Vector Machine (SVM).

Fig. 6. Comparative Analysis of the Proposed Face Detector and Existing Methods.

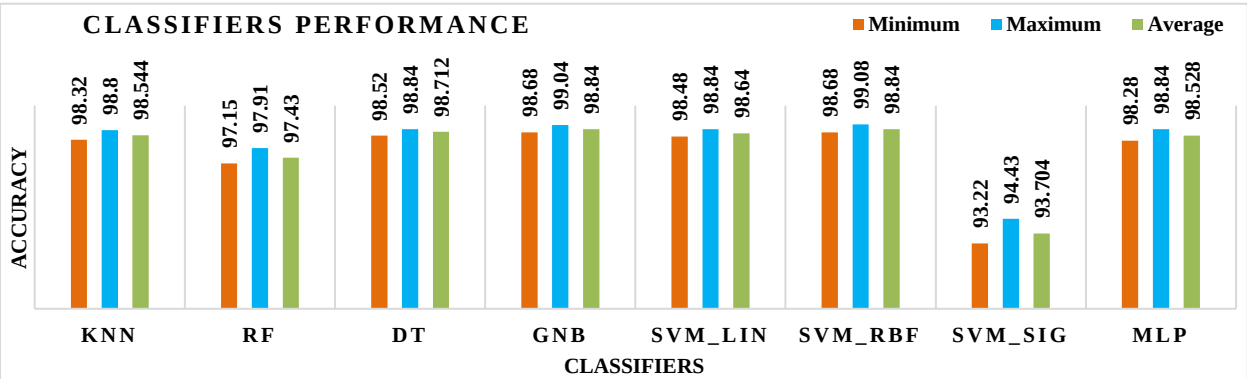
Table 3. Performance Analysis of the Proposed System.

Feature Extractor (No. of Features)	Accuracy	SVM Linear	SVM RBF	SVM Sigmoid
EfficientNetB0 (1002)	Minimum	79.68	78.89	15.58
	Maximum	81.02	80.14	17.89
	Average	80.49	79.34	17.19
EfficientNetB0 + LDA (05)	Minimum	90.62	90.71	77.19
	Maximum	91.10	91.47	78.77
	Average	90.83	91.04	77.82
HOG (1570)	Minimum	73.98	84.24	42.39
	Maximum	75.58	85.76	44.00
	Average	74.91	84.95	43.24
HOG + LDA (05)	Minimum	91.50	91.68	81.45
	Maximum	92.53	92.38	82.48
	Average	91.95	92.00	81.98
EfficientNetB0 + HOG (2570)	Minimum	82.52	87.85	64.92
	Maximum	84.00	89.78	66.12
	Average	83.22	88.80	65.60
EfficientNetB0 + HOG + LDA (05)	**Minimum**	**98.48**	**98.68**	**93.22**
	Maximum	**98.84**	**99.08**	**94.43**
	Average	**98.64**	**98.84**	**93.70**

Performance was evaluated across different kernel functions, with results summarised in Table 3 in terms of minimum, maximum, and average values, thereby demonstrating the robustness of the proposed synergistic feature-fusion framework. These findings provide the foundation for the subsequent discussion on recognition accuracy, comparative performance, and practical applicability in real-world scenarios. The experimental results indicate that the proposed combined model consistently outperforms approaches based on individual features as well as other alternative methods across all dataset settings. In particular, the feature space created by fusing both semantic and geometric descriptors demonstrated the most outstanding generalisation capability, especially when handling diverse sample variations. The consistent performance gains observed with incremental levels of feature fusion highlight the advantage of integrating deep learning features with handcrafted descriptors. In summary, the fusion of EfficientNetB0 features with HOG descriptors, optimised through LDA and classified using an SVM with an RBF kernel, provides a robust, efficient, and scalable solution for real-time recognition of children's facial expressions. These results underscore its strong applicability for deployment in practical emotion recognition systems.

5.2. Discussion

To establish a comprehensive benchmark, additional experiments were conducted using feature representations extracted from five widely used CNN architectures, AlexNet, GoogleNet, MobileNet, ResNet, and VGG19, alongside the handcrafted Local Binary Patterns (LBP) descriptor. Each feature set was evaluated both independently and within



the proposed two-level synergistic feature fusion framework, allowing assessment of the benefits of combining deep and handcrafted features. The detailed classification results are summarised in Table 4.

Fig. 7. Comparative analysis of classifiers.

Table 4. Comparative results with other combinations.

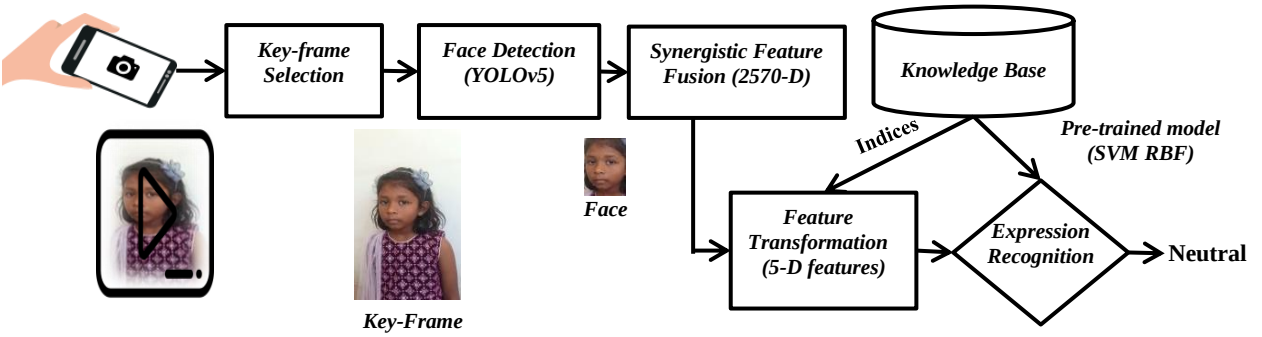
Model	SVM Linear	SVM RBF	SVM Sigmoid
AlexNet+EfficientNetb0	97.13	97.61	89.79
AlexNet+GoogleNet	96.66	96.88	89.84
AlexNet+HOG	97.88	98.32	92.07
AlexNet+LBP	86.97	87.47	73.82
AlexNet+MobileNet	96.06	96.37	87.88
AlexNet+ResNet	97.05	97.11	88.66
AlexNet+VGG19	95.29	95.57	88.07
EfficientNetb0+GoogleNet	97.30	97.49	91.12
EfficientNetb0+HOG	**98.64**	**98.84**	**93.70**
EfficientNetb0+LBP	91.84	92.29	80.58
EfficientNetb0+MobileNet	97.13	97.38	89.80
EfficientNetb0+ResNet	97.44	97.54	88.90
EfficientNetb0+VGG19	96.79	96.94	89.63
GoogleNet+HOG	98.19	98.41	92.78
GoogleNet+LBP	90.20	90.60	79.33
GoogleNet+MobileNet	96.87	97.10	89.65
GoogleNet+ResNet	97.02	97.20	89.74
GoogleNet+VGG19	96.37	96.48	89.78
HOG+LBP	93.16	93.46	82.48
HOG+MobileNet	98.50	98.56	92.92
HOG+ResNet	98.54	98.60	92.34
HOG+VGG19	98.01	98.20	92.79
LBP+MobileNet	88.55	89.18	77.64
LBP+ResNet	91.28	91.60	78.95
LBP+VGG19	83.10	83.42	71.57
MobileNet+ResNet	97.24	97.39	88.65
MobileNet+VGG19	95.49	95.97	87.85
ResNet+VGG19	96.69	96.91	88.70

The proposed framework was also evaluated across multiple classifiers, including K-Nearest Neighbour (KNN), Random Forest (RF), Decision Tree (DT), Gaussian Naïve Bayes (GNB), and Multilayer Perceptron (MLP). As illustrated in Figure 7, while most classifiers achieved moderate accuracy, the SVM with an RBF kernel consistently delivered the best performance, reaching 98.84% on the proposed dataset. Hyperparameter tuning further optimised the SVM-RBF classifier, with $C = 1$ and $\gamma = 1$ producing the highest accuracy. A direct comparison was also performed with three state-of-the-art facial expression recognition models, FER [30], DeepFace [29], and FEAT [28], on the developed children's facial dataset. The framework was tested under pose variations from 0° to 90°, recording minimum, average, and maximum performance metrics. Table 5 shows that while the existing models achieved average accuracies of 59.34%, 43.26%, and 49.13%, respectively, their maximum accuracies did not exceed 60.93%, highlighting their limitations in handling the variability and subtle expressions present in children's faces.

The significance of the proposed framework becomes even clearer when considering dimensionality reduction. Using the complete 2,570-dimensional feature set, the system achieved an accuracy of 88.80%. When reduced through LDA to only five highly discriminative features, accuracy improved markedly to 98.84%, demonstrating that LDA effectively compresses the feature space while enhancing class-relevant discriminative power. However, PCA alone achieved 87.19% accuracy with 476 features, and when combined with LDA, the accuracy further declined to 85.23%, illustrating that unsupervised dimensionality reduction is insufficient for preserving expression-specific information in this dataset. Overall, these results highlight the clear advantage of the proposed synergistic approach.

Table 5. Contemporary Analysis of Proposed System.

Facial Pose	Accuracy	Proposed (No. of Features)				Existing		
		Normal	LDA (5)	PCA (476)	PCA +	FEAT	DeepFace	FER



		(2570)			LDA (5)			
0^0 to 90^0	Minimum	87.85	**98.68**	85.20	83.63	57.84	41.60	47.75
	Maximum	89.78	**99.08**	88.31	86.18	60.93	44.20	50.38
	Average	88.80	**98.84**	87.19	85.23	59.34	43.26	49.13

Fig.8. Best-performing combination of the proposed framework.

By fusing deep features from EfficientNetB0 with handcrafted HOG descriptors, reducing dimensionality through LDA, and classifying with an optimised SVM-RBF, the framework consistently outperforms conventional methods and state-of-the-art models. This demonstrates its robustness, efficiency, and strong generalisation capability across diverse samples and pose variations, making it a highly effective solution for real-time children's facial expression recognition. The best-performing configuration is illustrated in Figure 8.

6. Conclusion and Future Work

This study presents a novel dataset of children's facial emotions, collected to capture diverse expressions, poses, and age groups, addressing a critical gap in existing resources. A key-frame selection method was designed to extract the most informative frames, while a customised YOLOv5 face detector ensured accurate localisation under varying conditions. The proposed framework leverages synergistic feature fusion, combining EfficientNetB0 deep features with HOG descriptors, and was additionally evaluated with features from other conventional (LBP, PCA) and deep learning models (AlexNet, GoogleNet, MobileNet, ResNet, VGG19). LDA was used to compress features into a small, highly discriminative set, thereby improving class separability. In contrast, SVM with an RBF kernel was chosen for its robustness in handling high-dimensional, non-linear decision boundaries, achieving an accuracy of 98.84%. The remaining 1.16% error is primarily due to subtle expressions and extreme pose variations. Compared to existing models such as FEAT, DeepFace, and FER, the framework demonstrates superior performance and robustness. Future work will expand the dataset to include larger age ranges and greater cross-cultural diversity, explore lightweight architectures for mobile and edge deployment, and investigate multimodal recognition and temporal sequence modelling using recurrent networks and transformers to further enhance real-time emotion recognition in educational and healthcare applications.

Acknowledgements

The first author of this work gratefully acknowledge Miss. Nithyashree D R and Miss. Kavya N for their support in creating dataset.

Data Availability

The dataset is available on request from the corresponding author.

Competing Interests

The authors declare that they have no conflicts of interest / competing interest towards this article.

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